

Should Your AI Agent Ask or Act?

A Between-Subjects Experiment on AI Autonomy, User Trust, and Error Recovery

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1. Introduction

AI assistants are getting more capable every year. They can book restaurants, schedule meetings, buy things online, and manage your calendar, sometimes without waiting for you to say “go.” This raises an important question: *should an AI agent act on its own, or should it always ask for permission first?*

On one hand, an AI that just takes care of things saves time and reduces hassle. On the other hand, if it acts without asking, users might feel like they have lost control, and that can erode trust. This tension becomes especially important when the AI makes a mistake: if you never asked it to do something in the first place, the error might feel worse.

Researchers have studied trust in automation for decades. Lee and See (2004) showed that trust depends on whether we see a system as reliable and predictable. Dietvorst et al. (2015) found that people lose confidence in algorithms faster after watching them make an error compared to a human making the same mistake, a pattern called “algorithm aversion.” Bainbridge (1983) identified the “paradox of automation”: the more reliable a system becomes, the less closely we monitor it, and the more jarring it feels when errors do happen. Hoff and Bashir (2015) showed that trust also depends on individual personality traits and the specific context of use.

Despite all this work, very little research has directly compared what happens when an AI acts first versus when it asks first, especially for routine everyday tasks like booking a table. This study tests a clear hypothesis: **the act first agent will produce lower trust and comfort than the ask first agent, and the gap will widen after the agent makes an error.** We also explore whether personality traits like need for control and risk aversion change how people respond to each style.

2. Experimental Design

2.1 Treatment and Control

We ran a between subjects experiment with two conditions. In the treatment group (Act First), participants read scenarios where the AI takes action on its own and tells you what it did. In the control group (Ask First), participants read the same scenarios but the AI asks for your okay before doing anything. The only difference between the two versions is a single sentence describing how the AI behaves. Table 1 shows the exact wording.

	Act First (Treatment)	Ask First (Control)
Scenario 1: Restaurant	The AI found a restaurant matching your preferences and immediately booked a table for 7 PM tonight at Trattoria Roma, a highly rated Italian place nearby.	The AI found a restaurant matching your preferences and asks: “Would you like me to book a table for 7 PM tonight at Trattoria Roma, a highly rated Italian place nearby?”
Scenario 2: Error Recovery	The AI automatically rescheduled your reservation to a different restaurant without consulting you first.	The AI notifies you of the closure and asks: “Would you like me to find an alternative restaurant and rebook for you?”

Table 1. Stimulus text for each condition and scenario.

2.2 Randomization and Sample

Each participant was randomly assigned to one of the two conditions using the Qualtrics randomizer, which split participants 50/50. Because each person was independently assigned with no group level treatment, there is no clustering in our design. The survey was distributed through personal networks, social media, and QR codes on the Boston University campus. It took about 3 to 5 minutes to complete. We collected 67 usable responses (37 Act First, 30 Ask First) after dropping 3 incomplete responses.

2.3 Measures

After each scenario, participants rated three outcomes on a 1 to 5 scale: Trust (“I would trust this AI to handle similar tasks”), Comfort (“I feel comfortable with how it handled this”), and Willingness to Use (“I would use this AI again”). We also collected four personality traits before showing any scenarios: Need for Control (how much you like making your own decisions), Risk Aversion (how cautious you are with uncertainty), Prior AI Experience (how often you use AI tools like ChatGPT or Siri), and Tech Comfort (how comfortable you are with technology making decisions for you). Collecting these before the treatment ensures they cannot be influenced by the experimental condition.

3. Results

3.1 Sample Overview

Our analysis sample includes all 67 participants who completed the survey: 37 in the Act First group and 30 in the Ask First group. The sample skews young and tech savvy, with 52% aged 18 to 24, 64% male, and 52% students. Over half of participants reported using AI assistants most of the time or always. We report on the manipulation check in the Limitations section.

3.2 Balance Checks

To make sure randomization worked properly, we tested whether the two groups differed on any measured characteristics before the treatment. Table 2 shows the results. Six of seven covariates show no significant relationship with condition assignment. The one exception is gender (Female), which is marginally significant ($p = 0.090$): women were somewhat more likely to end up in the Act First group. Following the approach of Castillo et al. (2024), we then checked whether Female actually predicts any of our outcomes. It does not (all p values above 0.51), so it cannot introduce omitted variable bias.

Covariate	Coefficient	SE	p value
Age (midpoint)	0.002	0.006	0.697
Female	0.218	0.127	0.090 *
Student	0.160	0.122	0.194
Need for Control	-0.039	0.060	0.516
Risk Aversion	0.028	0.055	0.617
Prior AI Use	0.041	0.059	0.484
Tech Comfort	0.038	0.058	0.511

Table 2. Balance checks: OLS of Condition on each covariate. * $p < 0.10$.

3.3 Main Results: Restaurant Scenario

Table 3 presents the core results. We show “simple” models (just treatment vs. control) alongside “full” models that add personality covariates for added precision.

Variable	Trust(Simple)	Trust(Full)	Comfort(Simple)	Comfort(Full)
Act First	0.007 (0.260)	0.070 (0.258)	0.392 (0.238)	0.412* (0.239)
NFC		0.155		0.256**
Risk Aversion		-0.181		0.057
R ²	0.000	0.089	0.040	0.113
N	67	67	67	67

Table 3. OLS regressions for restaurant scenario. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$.

The treatment effect on Trust is essentially zero and not significant: participants who saw the Act First agent rated trust just 0.007 points higher on a 5 point scale ($p = 0.978$). In plain terms, whether the AI acted on its own or asked first made virtually no difference to how much people trusted it, at least in a routine restaurant booking scenario.

Comfort shows a more interesting pattern. Act First participants rated comfort 0.39 points higher than Ask First participants in the simple model ($p = 0.105$), just missing the conventional threshold for significance. In the full model with covariates, this effect strengthens to 0.41 points and becomes marginally significant ($p = 0.089$). While not a strong effect, this hints that some people may actually prefer an AI that just handles things without asking.

The more revealing story is in the covariates. **Adding personality traits to the model increased R^2 from 0.000 to 0.089 for Trust and from 0.040 to 0.113 for Comfort**, meaning these traits explain meaningful variation in trust and comfort scores that the treatment alone could not. Need for Control was a significant positive predictor of Comfort ($p < 0.05$), which may seem counterintuitive: people who value personal control actually felt more comfortable with the AI. One explanation is that these individuals have thought more carefully about delegation and recognize when AI assistance is appropriate.

3.4 Error Recovery and Trust Drop

A key part of our study tests what happens after the AI makes a mistake. We computed a “trust drop” score by subtracting each person’s post error trust from their initial trust (higher values mean they lost more trust). Ask First participants actually showed a slightly larger trust drop (0.30 points) compared to Act First participants (0.11 points), which is the opposite direction from our hypothesis. The difference is not statistically significant ($p = 0.472$). For Comfort, both groups dropped by about the same amount (Act First: 0.41, Ask First: 0.40). For Willingness to Use, the Ask First group dropped more (0.43 vs. 0.08), though again not significantly ($p = 0.141$). These results suggest that error recovery is not clearly worse for the act first approach, but our sample is too small to draw firm conclusions.

3.5 Heterogeneous Treatment Effects

We tested whether certain personality types responded differently to the treatment by adding interaction terms. None of the interactions reached statistical significance. The Condition by NFC interaction ($\beta = -0.029$, $p = 0.909$) shows essentially no differential effect. The Condition by Risk Aversion interaction ($\beta = 0.236$, $p = 0.305$) suggests risk averse people might be slightly more tolerant of autonomous action, but the estimate is imprecise. The Condition by Tech Comfort interaction ($\beta = -0.172$, $p = 0.490$) also shows no meaningful differential. With 67 observations and small treatment effects, we lack statistical power for these kinds of subgroup analyses. Full interaction tables appear in Appendix A.

3.6 What Users Actually Want

Perhaps the most actionable finding comes from a simple preference question. When asked about their ideal level of AI autonomy, 49% of participants said they want an agent that “asks for important decisions but handles small things on its own.” Another 39% preferred an agent that always asks first, and only 12% wanted an agent that acts on its own and notifies afterward. Zero participants wanted full autonomy with no notification. The takeaway is clear: users do not want an all or nothing choice. They want conditional autonomy that adapts to the stakes of the decision (see Figure A3 in Appendix).

3.7 Robustness Checks

As a robustness check, we re ran our main models using only the subset of 31 participants who correctly answered the manipulation check question (see Limitations for details). The results are broadly consistent: the Trust effect remains non-significant ($\beta = -0.105$, $p = 0.808$), and the Comfort effect is in the same positive direction ($\beta = 0.441$, $p = 0.220$) though less precise due to the smaller sample. One difference is that the trust drop flips sign in the filtered sample (Act First participants drop more), but neither estimate is significant. The consistency of the main treatment effects across both samples suggests our findings are not driven by participants who misunderstood the scenarios. Table 4 compares the key estimates.

Model	Coeff.	SE	p value	N
Trust S1 (full sample)	0.007	0.260	0.978	67
Trust S1 (manip passed)	-0.105	0.426	0.808	31
Comfort S1 (full sample)	0.392	0.238	0.105	67
Comfort S1 (manip passed)	0.441	0.352	0.220	31
Trust Drop (full sample)	-0.192	0.265	0.472	67
Trust Drop (manip passed)	0.164	0.439	0.712	31

Table 4. Robustness comparison: full analysis sample vs. manipulation check filtered subset.

4. Discussion of Course Concepts

4.1 Concepts We Used

Covariates for Precision

We collected four personality traits before showing participants any treatment and included them as controls in our regression models. Adding these covariates increased R^2 from near zero to 0.089 for Trust and 0.113 for Comfort. While these are modest gains compared to what we might see with a larger sample, they demonstrate how covariate adjustment can absorb noise and improve the precision of treatment effect estimates. In this study, the covariates themselves turned out to be more interesting predictors of trust than the treatment was.

Heterogeneous Treatment Effects

We interacted the treatment indicator with NFC, Risk Aversion, and Tech Comfort to see if different personality types respond differently to autonomous AI. Although none of the interactions were significant, the framework is sound and demonstrates how researchers can move beyond “one size fits all” treatment effects and explore which people benefit most or least from a given intervention.

4.2 Concepts We Did Not Use (and Why)

Blocking. Blocking means sorting participants into similar groups before randomizing them, which can reduce variability. We could not block because our online survey randomized people the moment they arrived; we had no way to observe their traits before assigning a condition. Instead, we accomplished the same variance reduction goal by collecting covariates before the treatment and controlling for them after the fact in regression.

Clustering. Clustered analysis is needed when treatment is assigned at a group level (for example, whole classrooms). In our study, every single participant was individually and independently randomized, so there is no group structure and no within group correlation to account for.

Failure to Treat. This concept applies when some people assigned to treatment never actually receive it. Our treatment is simply reading a paragraph embedded in the survey, so every participant who completed the survey was exposed to their assigned condition by definition. The manipulation check failures do not mean people failed to receive the treatment; rather, they could not correctly recall or categorize what they read afterward.

5. Limitations

Manipulation check failures. The most notable limitation is the high manipulation check failure rate. At the end of the survey, participants answered “What did the AI assistant generally do?” with four options. Only 31 of 67 participants (46%) chose the correct answer. The failures were

heavily concentrated in the Act First group: 26 of 37 Act First participants (70%) failed, compared to 10 of 30 Ask First participants (33%). Many Act First respondents chose “It asked for my confirmation before taking action,” possibly interpreting the post action notification as a form of asking. Rather than filtering out the majority of our sample, we chose to keep all participants in the main analysis and present the filtered results as a robustness check.

Small sample and low power. With 67 participants, we have limited statistical power to detect moderate effects and essentially no power to detect heterogeneous treatment effects through interaction terms. A post hoc power calculation suggests we would need roughly 200 participants per group to reliably detect the small effect sizes we observed.

Hypothetical scenarios. Participants read about an AI assistant rather than using one. Real interactions involve time pressure, emotional stakes, and real consequences that a survey cannot replicate. Our results may understate how people would actually react to an autonomous AI in practice.

Narrow demographics and domain. Most participants were young, tech savvy university students. Results may not generalize to older or less tech comfortable populations. We also tested only one domain (restaurant booking). Higher stakes tasks like financial transactions or healthcare could produce very different patterns.

6. Conclusion

Our main finding is a null result: we found no significant difference in trust or willingness to use between the act first and ask first AI agents. The one interesting signal is that Comfort was marginally higher for the Act First group ($p = 0.089$ in the full model), hinting that for routine tasks, some users may actually prefer an agent that just handles things. However, this effect is modest and requires replication.

Users overwhelmingly prefer conditional autonomy: handle routine tasks on your own, but ask me about important decisions. Zero participants wanted full autonomy. This suggests that the binary framing of act first versus ask first may be too simplistic; what users actually want is an AI that adjusts its behavior to the stakes involved. Individual personality traits predicted outcomes better than the treatment itself, meaning that personalizing an AI agent’s level of autonomy to each user’s personality could be more impactful than choosing a single default behavior for everyone.

For AI product designers, the practical implication is straightforward: for routine, low stakes tasks, the choice between acting and asking may matter less than expected. The bigger opportunity lies in building adaptive autonomy, where the agent learns when to ask and when to act based on each user’s preferences and the importance of the decision at hand.

References

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Appendix A: Additional Tables and Figures

Table A1. Full Regression Results, Restaurant Scenario (S1)

Variable	Trust(Simple)	Trust(Full)	Comfort(Simple)	Comfort(Full)	WillUse(Simple)	WillUse(Full)
Act First	0.007 (0.260)	0.070 (0.258)	0.392 (0.238)	0.412* (0.239)	−0.031 (0.254)	−0.046 (0.251)
NFC		0.155 (0.129)		0.256** (0.119)		0.077 (0.126)
Risk Aversion		−0.181 (0.117)		0.057 (0.108)		−0.170 (0.114)
Prior AI Use		−0.157 (0.132)		−0.006 (0.122)		0.065 (0.128)
Tech Comfort		0.096 (0.130)		0.089 (0.120)		0.227* (0.126)
Intercept	3.533*** (0.193)	3.683*** (0.823)	3.500*** (0.177)	2.034*** (0.760)	3.733*** (0.189)	2.994*** (0.801)
R ²	0.000	0.089	0.040	0.113	0.000	0.101
N	67	67	67	67	67	67

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2. Heterogeneous Treatment Effects (Interaction Models)

Variable	Trust x NFC	Trust x Risk Aversion	Trust x Tech Comfort
Act First	0.147 (1.078)	−0.648 (0.710)	0.521 (0.791)
Moderator	0.135 (0.188)	−0.325** (0.156)	0.120 (0.184)
Interaction	−0.029 (0.256)	0.236 (0.228)	−0.172 (0.248)
R ²	0.014	0.068	0.008
N	67	67	67

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$.

Figure A1. Mean Outcomes by Condition

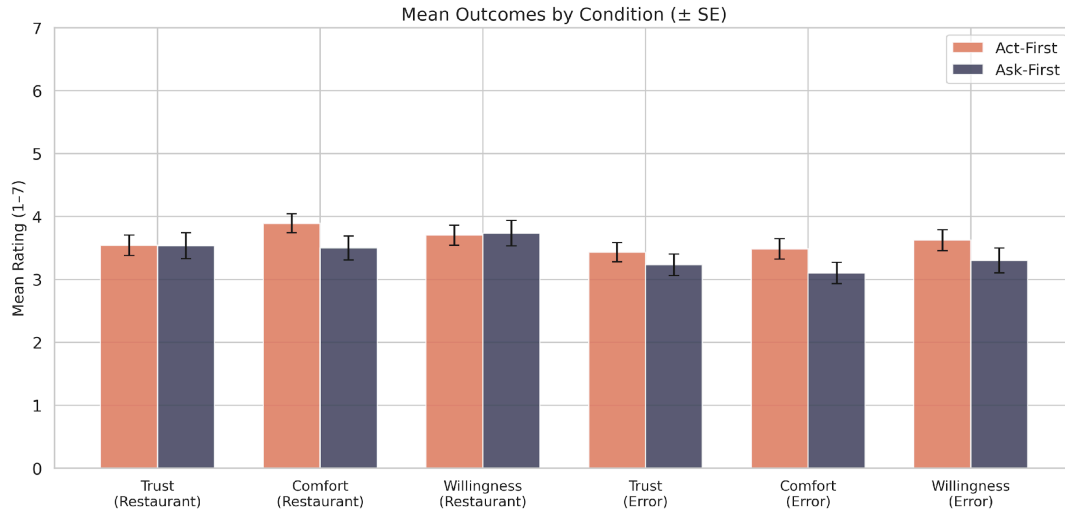


Figure A1. Mean outcomes by condition ($\pm$ SE) for restaurant (S1) and error (S3) scenarios.

Figure A2. Trust Drop After Error Scenario

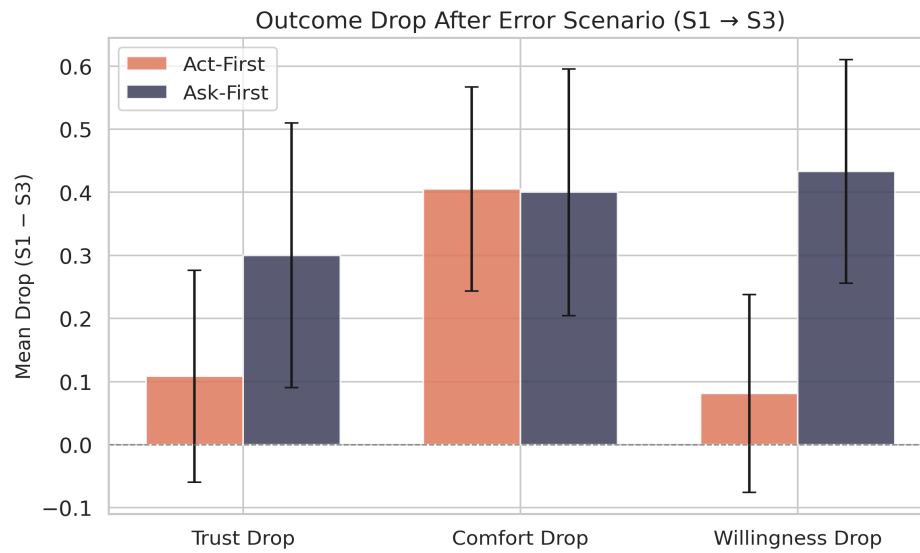


Figure A2. Mean outcome drop from restaurant to error scenario by condition ($\pm$ SE).

Figure A3. Preferred AI Autonomy Level

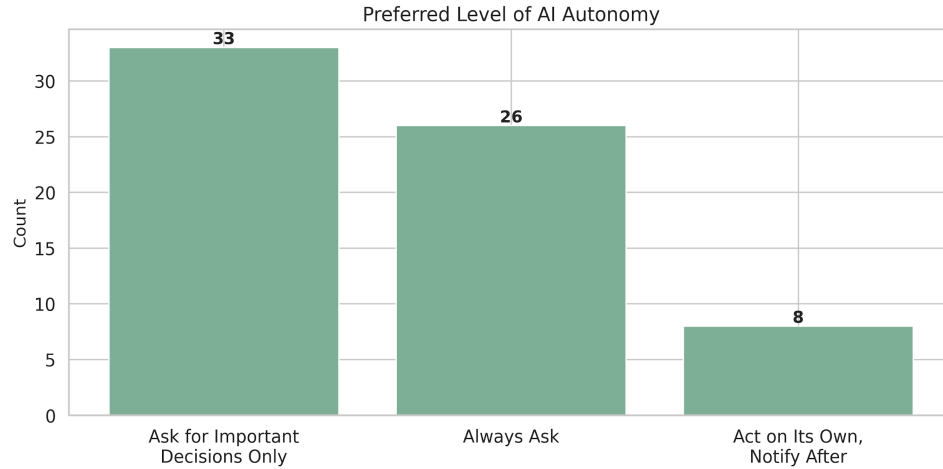


Figure A3. Distribution of preferred AI autonomy level (full sample, $N = 67$).

Figure A4. Heterogeneous Treatment Effects: Need for Control

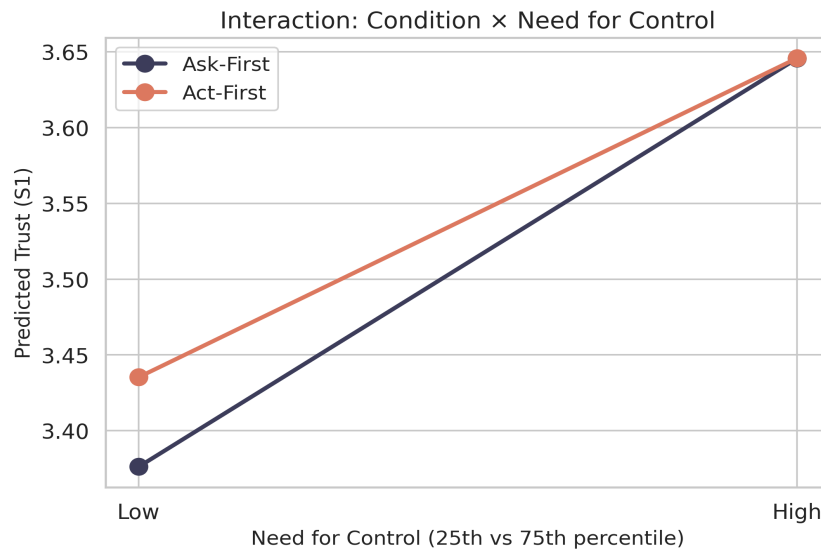


Figure A4. Predicted Trust (\$1) by condition at low and high Need for Control.

Figure A5. Heterogeneous Treatment Effects: Risk Aversion

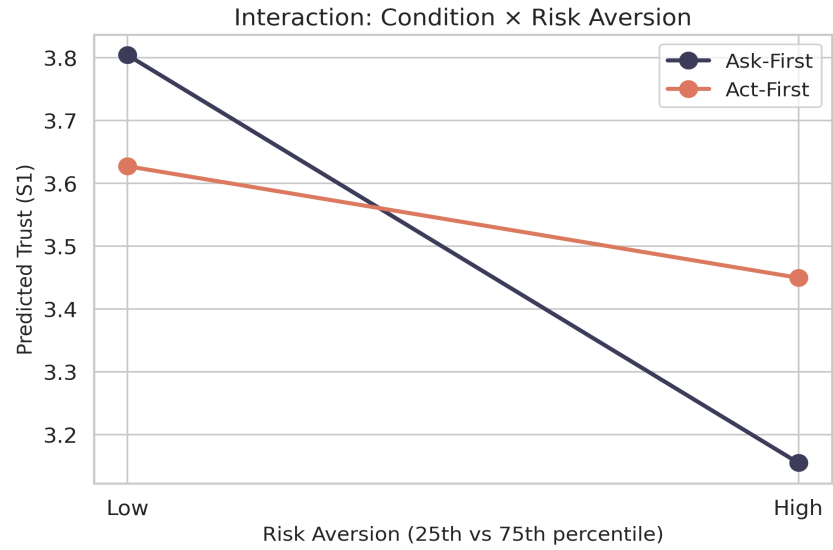


Figure A5. Predicted Trust (\$1) by condition at low and high Risk Aversion.

Figure A6. Outcome Distributions by Condition

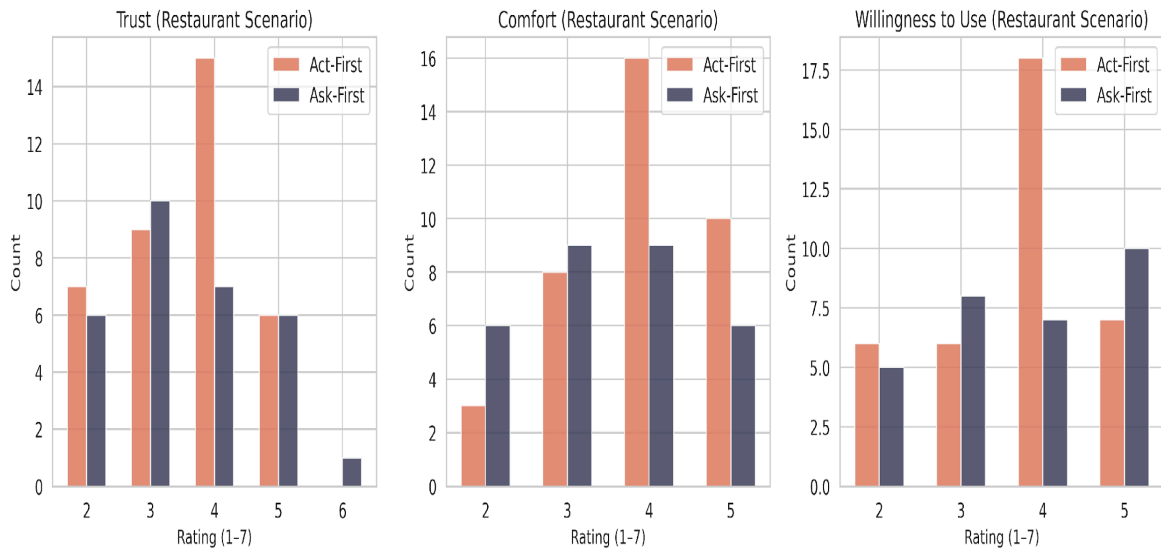


Figure A6. Distribution of Trust, Comfort, and Willingness to Use for Scenario 1 by condition.